

THEORY AND METHODS
THREE METHODS TO APPLY THE
BOOTSTRAP CORRECTLY TO HYPOTHESIS
TESTING

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Summary: In this paper we discuss three methods to apply the bootstrap correctly to hypothesis testing. For each of these methods a simple example is considered, namely testing the variance of a population. The shortcoming associated with implementing bootstrap testing procedures incorrectly is analysed by making use of this example. Some theoretical properties of the corresponding bootstrap critical values are derived. The results of a simulation study are presented to provide empirical comparison between the different methods and to substantiate the theoretical findings.

1. Introduction

The construction of bootstrap confidence intervals are linked to the execution of bootstrap hypothesis tests because of the duality between confidence intervals and hypothesis tests, i.e. the null hypothesis is rejected if the hypothesized value under the null hypothesis lies outside the confidence interval. Shao and Tu (1995), however, provided two reasons why it is important to consider bootstrap hypothesis testing separately: “*Firstly, finding a test directly is much easier than getting a test through constructing a confidence interval, which is*

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impossible in some cases. Secondly, the tests obtained directly may be better since they usually take account of the special nature of the hypothesis."

Davidson and MacKinnon (2004) also argue that there are many more ways to construct bootstrap confidence intervals (e.g. percentile intervals, etc.) than there are to perform bootstrap-based tests. As a result, these alternative ways to compute bootstrap confidence intervals may lead to different results and can thus be confusing.

Although there is a very large amount of literature on the bootstrap, only a small proportion of it is devoted to bootstrap-based testing. The focus, however, is usually on deriving bootstrap standard errors and constructing bootstrap confidence intervals. Hall and Wilson (1991) highlighted two important guidelines for bootstrap hypothesis testing. Their first guideline states that, when one wants to estimate the critical value, resampling must be done in a way that reflects the null hypothesis. This must be done even if the data were generated from a distribution specified by the alternative hypothesis. This guideline is crucial to the success of bootstrap hypothesis testing and has been mentioned by Beran (1988); Hinkley (1988); Fisher and Hall (1990); Westfall and Young (1993) and recently by Martin (2007). In their second guideline, Hall and Wilson (1991) recommended that bootstrap hypothesis tests should be based on test statistics that are (asymptotically) pivotal. The importance of using pivotal statistics in the bootstrap was considered by, among others, Beran (1988) and Hall (1992).

In this paper we shall consider three methods to apply the bootstrap correctly to hypothesis testing. In each case the method will be described, followed by an example, namely testing the variance of a population. The shortcoming associated with implementing bootstrap testing procedures

incorrectly is also analysed by making use of this example. The content of the article will be arranged in the following way: Section 2 introduces the transformation method to apply the bootstrap to hypothesis testing. In Section 3 we discuss the exponentially tilted version of the empirical distribution function (e.d.f.) followed by a discussion on resampling residuals in Section 4. The paper concludes with the results of a Monte Carlo study and some final remarks.

2. Transformation method

Let $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ be a sample of independent and identically distributed (i.i.d.) random variables with unknown distribution function (d.f.) $F \equiv F_{\boldsymbol{\theta}, \boldsymbol{\tau}}$, depending on unknown parameter vectors $\boldsymbol{\theta} = \boldsymbol{\theta}(F)$ and $\boldsymbol{\tau} = \boldsymbol{\tau}(F)$.

Consider the hypothesis

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \quad \text{vs.} \quad H_A : \boldsymbol{\theta} \in \Theta_A, \quad (1)$$

where Θ_0 and Θ_A are two disjoint subsets of some parameter space $\Theta = \Theta_0 \cup \Theta_A$. For testing H_0 we consider $\boldsymbol{\tau}$ as a nuisance parameter, which will be replaced by an estimator that is strongly consistent under both H_0 and H_A . These estimated nuisance parameters will therefore not be indicated in the notation of test statistics and critical values.

Assume, without loss of generality, that the test procedure is of the form:

Reject H_0 if and only if

$$T_n(\mathbf{X}_n) \geq C_n(\alpha; \mathbf{X}_n),$$

where $T_n(\mathbf{X}_n)$ is an appropriate test statistic and $C_n(\alpha; \mathbf{X}_n)$ is a bootstrap critical value, which depends on the significance level α ($0 < \alpha < 1$) of the test and the given set of data $\mathbf{X}_n$. $C_n(\alpha; \mathbf{X}_n)$ is defined as follows:

Let $\mathbf{X}_n^* = (X_1^*, X_2^*, \dots, X_n^*)$ be a bootstrap random sample drawn with replacement from the e.d.f. F_n of $\mathbf{X}_n$. In the bootstrap world (for fixed $\mathbf{X}_n$) we would like to have that $\theta(F_n) \in \Theta_0$. However, this is seldom if ever the case, hence we need to transform the data $X_1, X_2, \dots, X_n$. Denote the transformed variables by

$$W_i^0 = W_i^0(\mathbf{X}_n), \quad i = 1, 2, \dots, n.$$

These variables are chosen such that $\theta(G_n) \in \Theta_0$ in the bootstrap world, where G_n is the e.d.f. of $\mathbf{W}_n^0 = (W_1^0, W_2^0, \dots, W_n^0)$. The bootstrap random sample is given by $\mathbf{W}_n^{0*} = (W_1^{0*}, W_2^{0*}, \dots, W_n^{0*})$ drawn with replacement from G_n . Usually, $\mathbf{W}_n^{0*}$ will depend on both $\mathbf{X}_n$ and $\mathbf{X}_n^*$. The bootstrap estimator $C_n(\alpha; \mathbf{X}_n)$ is now chosen such that

$$P^*(T_n(\mathbf{W}_n^{0*}) \geq C_n(\alpha; \mathbf{X}_n)) \cong \alpha,$$

where P^* denotes the conditional probability law of $\mathbf{W}_n^{0*}$ given $\mathbf{X}_n$. Note that $C_n(\alpha; \mathbf{X}_n)$ can be approximated by a simple Monte Carlo algorithm suggested by Efron (1979). Throughout the paper this approximation will be denoted by $\hat{C}_n(\alpha; \mathbf{X}_n)$.

Example 1 Testing the variance of a population.

In this example we consider a bootstrap test for the hypothesis

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_A : \sigma^2 > \sigma_0^2,$$

based on a random sample $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ obtained from an unknown d.f. F with finite mean μ and variance σ^2 , $0 < \sigma^2 < \infty$.

Possible test statistics for this scenario are

$$T_{n,N-P}(\mathbf{X}_n) = \frac{nS_n^2(\mathbf{X}_n)}{\sigma_0^2} \quad \text{and} \quad T_{n,P}(\mathbf{X}_n) = \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}},$$

where

$$S_n^2(\mathbf{X}_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2; \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad \hat{\mu}_4(\mathbf{X}_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^4.$$

The subscripts P and $N - P$ in the above notation refer to a pivotal (or asymptotically pivotal) and non-pivotal statistic, respectively. Note that, even though the statistic $T_{n,N-P}(\mathbf{X}_n) = nS_n^2(\mathbf{X}_n)/\sigma_0^2$ is a pivot under normality, it is non-pivotal for other distributions. In the remainder of the paper, we shall only be using the pivotal test statistic to illustrate some concepts. However, in the Monte Carlo study we will compare the performance of bootstrap-based tests based on both the pivotal and non-pivotal statistics.

Consider the following transformation of $\{X_i, i = 1, 2, \dots, n\}$:

$$W_i^0 = \left(\frac{\sigma_0}{S_n(\mathbf{X}_n)} \right) X_i, \quad i = 1, 2, \dots, n.$$

By construction, the sample variance of $\{W_i^0, i = 1, 2, \dots, n\}$ is σ_0^2 . The bootstrap critical value $C_{n,P}^R(\alpha; \mathbf{X}_n)$ is now defined by

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{W}_n^{0*}) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{W}_n^{0*}) - S_n^4(\mathbf{W}_n^{0*})}} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha,$$

where

$$\begin{aligned} S_n^2(\mathbf{W}_n^{0*}) &= \frac{1}{n} \sum_{i=1}^n (W_i^{0*} - \bar{W}_n^{0*})^2; \quad \bar{W}_n^{0*} = \frac{1}{n} \sum_{i=1}^n W_i^{0*} \\ \text{and} \quad \hat{\mu}_4(\mathbf{W}_n^{0*}) &= \frac{1}{n} \sum_{i=1}^n (W_i^{0*} - \bar{W}_n^{0*})^4. \end{aligned}$$

Because of the transformation, it easily follows that $C_{n,P}^R(\alpha; \mathbf{X}_n)$ can also be computed from

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - S_n^2(\mathbf{X}_n))}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha, \quad (2)$$

where

$$S_n^2(\mathbf{X}_n^*) = \frac{1}{n} \sum_{i=1}^n (X_i^* - \bar{X}_n^*)^2; \quad \bar{X}_n^* = \frac{1}{n} \sum_{i=1}^n X_i^* \quad \text{and} \quad \hat{\mu}_4(\mathbf{X}_n^*) = \frac{1}{n} \sum_{i=1}^n (X_i^* - \bar{X}_n^*)^4.$$

If, however, the data are not transformed to have variance σ_0^2 , the bootstrap critical value is defined by

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \right) \cong \alpha. \quad (3)$$

The symbol R in the notation of $C_{n,P}^R(\alpha; \mathbf{X}_n)$ is used to denote the “right” critical value, obtained when the resampling is done from the transformed data $\mathbf{W}_n^0$ in order to “mimic” H_0 in the bootstrap world, whereas the W in the notation of $C_{n,P}^W(\alpha; \mathbf{X}_n)$ is used to indicate the “wrong” critical value, derived when the bootstrap is applied wrongly, i.e. the resampling is done from the original data $\mathbf{X}_n$, and thus does not mimic H_0 . Again, the subscripts P and $N - P$ will refer to the pivotal and non-pivotal cases, respectively.

The following two theorems describe the behaviour of $C_{n,P}^R(\alpha; \mathbf{X}_n)$ and $C_{n,P}^W(\alpha; \mathbf{X}_n)$ with respect to size properties of the test.

Theorem 1 If $E(X_1^8) < \infty$ then, as $n \rightarrow \infty$,

$$P_{H_0} \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \right) \rightarrow \begin{cases} 1, & \text{when } \alpha \geq 0.5 \\ 0, & \text{when } \alpha < 0.5. \end{cases}$$

Proof.

Clearly, from (3) we have that

$$\begin{aligned} & P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \right) \cong \alpha \\ \Leftrightarrow & P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - S_n^2(\mathbf{X}_n))}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} + \frac{\Delta_n^*}{\sqrt{n}} \leq C_{n,P}^W(\alpha; \mathbf{X}_n) \right. \\ & \quad \left. - \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \right) \cong 1 - \alpha, \quad (4) \end{aligned}$$

where

$$\Delta_n^* = \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \cdot \frac{\sqrt{n} \left(\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)} - \sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)} \right)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} =: \Delta_{n1} \cdot \Delta_{n2}^*.$$

The CLT implies that, under H_0 , $\Delta_{n1} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$, so that $\Delta_{n1} = O_P(1)$. From the equation given in (27.5.4) of Cramér (1945) it follows that

$$\text{Var}(\hat{\mu}_4(\mathbf{X}_n)) = \frac{\sigma_1^2}{n} + O\left(\frac{1}{n^2}\right), \quad (5)$$

where

$$\sigma_1^2 = \mu_8 - 8\mu_3\mu_5 - \mu_4^2 + 16\mu_2\mu_3^2,$$

and

$$\mu_\nu = E((X_1 - \mu)^\nu), \quad \nu = 2, 3, \dots$$

Now, from (5) and by applying classical bootstrap theory it readily follows, using the assumption $E(X_1^8) < \infty$, that

$$\Delta_{n2}^* \xrightarrow{d^*} N(0, \sigma_1^2 / (4(\mu_4 - \sigma^4)^2)) \quad \text{a.s. as } n \rightarrow \infty, \quad (6)$$

where d^* denotes convergence in conditional, given $\mathbf{X}_n$, distribution (i.e. weak convergence in the bootstrap world).

Let $a_n = n^{-1/2} \log n$. Then, using the fact that $\Delta_{n1} = O_P(1)$ and (6) we have that

$$\begin{aligned} P^*(n^{-1/2} |\Delta_n^*| > a_n) &= P^*(|\Delta_{n1}| \cdot |\Delta_{n2}^*| > \log n) \\ &= o_{a.s.}(1) \quad \text{as } n \rightarrow \infty. \end{aligned} \quad (7)$$

From (4), (7) and the Lemma given on pp. 228 – 229 by Serfling (1980) we therefore have that

$$\Phi \left(C_{n,P}^W(\alpha; \mathbf{X}_n) - \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \right) = 1 - \alpha + o_{a.s.}(1),$$

implying that

$$C_{n,P}^W(\alpha; \mathbf{X}_n) = \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} + \Phi^{-1}(1 - \alpha) + o_{a.s.}(1), \quad (8)$$

with Φ the standard normal d.f.

Hence, from (8) and Slutsky's theorem, it follows that

$$\begin{aligned} & P_{H_0} \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \right) \\ &= P_{H_0} \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \geq \frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} + \Phi^{-1}(1 - \alpha) + o_{a.s.}(1) \right) \\ &= P_{H_0} (0 \geq \Phi^{-1}(1 - \alpha) + o_{a.s.}(1)) \\ &\rightarrow I(0 \geq \Phi^{-1}(1 - \alpha)) \text{ a.s. as } n \rightarrow \infty, \end{aligned} \quad (9)$$

where $I(\cdot)$ is the indicator function. The proof of the theorem now follows from (9) and the fact that $\Phi^{-1}(1 - \alpha) > 0$ if $\alpha < 1/2$ and $\Phi^{-1}(1 - \alpha) \leq 0$ if $\alpha \geq 1/2$. ■

Theorem 2 Suppose $E(X_1^8) < \infty$ and that Cramér's continuity condition holds, i.e.

$$\limsup_{|t| \rightarrow \infty} |\chi(t)| < 1,$$

where $\chi(t)$ is the characteristic function of X_1 , then as $n \rightarrow \infty$,

$$P_{H_0} \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) = \alpha + O(n^{-1}).$$

Proof.

Under $H_0 : \sigma^2 = \sigma_0^2$, the Studentized statistic

$$\frac{\sqrt{n}(S_n^2(\mathbf{X}_n) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n) - S_n^4(\mathbf{X}_n)}}$$

belongs to a class of statistics defined by the so-called “smooth function model” (e.g. see Chapter 2, Section 2.4, of Hall, 1992).

Furthermore, Hall (1992) defines a second-order correct one-sided confidence interval $\hat{J}_1$ (given in (3.10) of his book) for the parameter associated with the smooth function model (in our case the variance σ^2). On page 103 Hall (1992) proves that it has coverage error of order n^{-1} .

The proof of our theorem now follows immediately from the duality between the confidence bound $\hat{J}_1$ and our one-sided hypothesis test, viz. H_0 is rejected if and only if σ_0^2 lies outside $\hat{J}_1$. ■

3. Exponentially tilted version of the e.d.f.

In Section 2 we discussed the transformation method to transform the data to mimic H_0 in the bootstrap world. An alternative approach suggested by Efron (1981) is to leave the data values fixed and instead change the probability (p_i) on each data value X_i to mimic H_0 in the bootstrap world. To be more specific, consider the hypothesis in (1):

Let $\mathbf{p} = (p_1, p_2, \dots, p_n)$ and let F_{Tilt} be the d.f. which places probability p_i on each X_i , $i = 1, 2, \dots, n$. It is now possible to choose $\mathbf{p}$ in such a way that the distance between F_{Tilt} and F_n is minimised, subject to the constraints:

$$\theta(F_{Tilt}) \in \Theta_0 \text{ (in the bootstrap world)} \quad \text{and} \quad \sum_{i=1}^n p_i = 1.$$

Denote this solution by $\mathbf{p}^0 = (p_1^0, p_2^0, \dots, p_n^0)$. A useful distance measure for this constrained minimisation problem is the reverse information distance or *Kullback-Leibler* distance

$$\sum_{i=1}^n p_i \log(np_i).$$

Alternative measures suggested by Young (1988) and Davison and Hinkley (1997) that can be used include maximising $\prod_{i=1}^n p_i$ (i.e. minimising $\sum_{i=1}^n \frac{1}{n} \log \left(\frac{1}{np_i} \right)$) or minimising $\max_{1 \leq i \leq n} |p_i - \frac{1}{n}|$.

After obtaining $\mathbf{p}^0$, the solution to the constrained minimisation problem, the bootstrap random sample $\mathbf{V}_n^{0*} = (V_1^{0*}, V_2^{0*}, \dots, V_n^{0*})$ drawn with replacement from F_{Tilt}^0 (i.e. F_{Tilt} based on $\mathbf{p}^0$), can be obtained. The bootstrap critical value $C_n^T(\alpha; \mathbf{X}_n)$ is then defined by

$$P^* (T_n(\mathbf{V}_n^{0*}) \geq C_n^T(\alpha; \mathbf{X}_n)) \cong \alpha,$$

where the T in the notation of $C_n^T(\alpha; \mathbf{X}_n)$ refers to the tilted bootstrap.

Efron (1981) and Young (1988) used the tilted bootstrap to test for the mean of a population, while Davison and Hinkley (1997) discussed the use of the tilted bootstrap when testing for the difference between two population means. We shall now again consider testing the variance of a population using the tilted bootstrap.

Example 1 (Continued)

In order to obtain $\mathbf{p}^0$ we minimise the Lagrange expression

$$L(\mathbf{p}, \lambda) = \sum_{i=1}^n p_i \log(np_i) - \lambda \left(\sum_{i=1}^n p_i (X_i - \bar{X}_p)^2 - \sigma_0^2 \right),$$

where $\sum_{i=1}^n p_i (X_i - \bar{X}_p)^2$ is the variance of F_{Tilt} , with $\bar{X}_p = \sum_{i=1}^n p_i X_i$.

Setting the derivative with respect to p_i equal to zero, produces the following equality:

$$\log(np_i) + 1 - \lambda(X_i - \bar{X}_p)^2 = 0,$$

which together with the constraint $\sum_{i=1}^n p_i = 1$, gives the implicit solution

$$p_i^0 = \frac{e^{\lambda(X_i - \bar{X}_p^0)^2}}{\sum_{j=1}^n e^{\lambda(X_j - \bar{X}_p^0)^2}}. \quad (10)$$

If we assume that $\bar{X}_p^0 = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, (10) becomes

$$p_i^0 = \frac{e^{\lambda(X_i - \bar{X})^2}}{\sum_{j=1}^n e^{\lambda(X_j - \bar{X})^2}},$$

where λ is chosen such that $\sum_{i=1}^n p_i^0 (X_i - \bar{X})^2 = \sigma_0^2$. That is, λ is chosen such that

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2 e^{\lambda(X_i - \bar{X})^2}}{\sum_{j=1}^n e^{\lambda(X_j - \bar{X})^2}} = \sigma_0^2.$$

4. Resampling residuals

When working with complicated data sets (like regression models), transforming the data in order to mimic H_0 (or H_A) in the bootstrap world can be very difficult. An alternative way is to resample residuals, and then make use of the resampled residuals to build a model that conforms to H_0 (or H_A). Examples of where this technique is applied to perform bootstrap-based testing are especially common in the econometric literature. Some of these applications include: bootstrapping p-values and power in the first-order autoregression model (Rayner, 1990); the level and power of the bootstrap t -test in the AR(1)-model with trend (Nankervis and Savin, 1996); bootstrap testing in nonlinear models (Davidson and MacKinnon, 1999); implementing the bootstrap in static or stable dynamic regression models (van Giesbergen and Kiviet, 2002); bootstrap unit root tests (Park, 2003) and testing for autoregressive conditional heteroscedasticity (Davidson and MacKinnon, 2007).

We shall first provide an algorithm of how to resample residuals in order to correctly apply the bootstrap to hypothesis testing, followed by an example illustrating how important it is that residuals must be resampled in a correct manner.

Consider data $((\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \dots, (\mathbf{X}_n, Y_n))$. Suppose we are interested in testing hypotheses concerning parameters from the following model:

$$Y_t = G(\mathbf{X}_t, \boldsymbol{\beta}, \mathbf{Y}_t^L, \boldsymbol{\phi}) + \varepsilon_t, \quad t = p+1, p+2, \dots, n,$$

where

- $\mathbf{Y}_n = (Y_1, Y_2, \dots, Y_n)^T$ are the response variables;
- $\{\varepsilon_t\}$ are i.i.d. random errors from an unknown d.f. F , such that $E(\varepsilon_t) = 0$ and $E(\varepsilon_t^2) = \sigma^2$;
- $\mathbf{X}_t = (X_{t,1}, X_{t,2}, \dots, X_{t,k+m})$ are regressors (which can be fixed or random);
- $\boldsymbol{\beta} = (\boldsymbol{\theta}, \boldsymbol{\nu}) \in \Pi = \Theta \times N \subset R^k \times R^m$ is a set of parameters associated with $\mathbf{X}_t$;
- $\mathbf{Y}_t^L = (Y_{t-1}, Y_{t-2}, \dots, Y_{t-p})$ are the lagged values of the response variables;
- $\boldsymbol{\phi} = (\boldsymbol{\rho}, \boldsymbol{\gamma}) \in \Omega = P \times \Gamma \subset R^q \times R^s$ is a set of parameters associated with $\mathbf{Y}_t^L$, $q + s = p$; and
- G is some known function.

Let $\boldsymbol{\theta}$ be the parameter of interest and suppose we wish to test the hypothesis:

$$H_0 : \mathbf{r}'\boldsymbol{\theta} = \mathbf{r}'\boldsymbol{\theta}_0 \quad \text{vs.} \quad H_A : \mathbf{r}'\boldsymbol{\theta} > \mathbf{r}'\boldsymbol{\theta}_0,$$

where $\mathbf{r}$ is a known k -dimensional column vector and $\boldsymbol{\nu}, \boldsymbol{\rho}$ and $\boldsymbol{\gamma}$ are a set of m, q and s nuisance parameters, respectively.

The test rejects H_0 if and only if

$$T_n(\mathbf{Y}_n, \boldsymbol{\chi}_n) \geq C_n(\alpha; \mathbf{Y}_n, \boldsymbol{\chi}_n),$$

where $T_n(\mathbf{Y}_n, \boldsymbol{\chi}_n)$ is an appropriate test statistic, $C_n(\alpha; \mathbf{Y}_n, \boldsymbol{\chi}_n)$ is a bootstrap critical value and $\boldsymbol{\chi}_n = (\mathbf{X}_{p+1}, \mathbf{X}_{p+2}, \dots, \mathbf{X}_n)$. The following residual-based bootstrap algorithm can be used to obtain $C_n(\alpha; \mathbf{Y}_n, \boldsymbol{\chi}_n)$:

- (i) Let $\hat{\boldsymbol{\beta}}_n = (\hat{\boldsymbol{\theta}}_n, \hat{\boldsymbol{\nu}}_n) \in \Pi$ and $\hat{\boldsymbol{\phi}}_n = (\hat{\boldsymbol{\rho}}_n, \hat{\boldsymbol{\gamma}}_n) \in \Omega$ be estimators for $\boldsymbol{\beta}$ and

ϕ , respectively.

- (ii) Define centred residuals, for $t = p + 1, p + 2, \dots, n$, by

$$\hat{\varepsilon}_t = Y_t - G(\mathbf{X}_t, \hat{\beta}_n, \mathbf{Y}_t^L, \hat{\phi}_n) - \frac{1}{n-p} \sum_{j=p+1}^n (Y_j - G(\mathbf{X}_j, \hat{\beta}_n, \mathbf{Y}_j^L, \hat{\phi}_n)) \quad (11)$$

- (iii) Under H_0 , Y_t can be represented as

$$W_t^0 = G(\mathbf{X}_t, \tilde{\beta}_0, \mathbf{W}_t^{L0}, \phi) + \varepsilon_t^0, \quad t = p + 1, p + 2, \dots, n,$$

where

$$\tilde{\beta}_0 = (\theta_0, \nu); \quad \mathbf{W}_t^{L0} = (W_{t-1}^0, W_{t-2}^0, \dots, W_{t-p}^0),$$

and $\{\varepsilon_t^0\}$ are unknown error terms under H_0 .

- (iv) Under H_0^* (the bootstrap null hypothesis), W_t^0 can be represented as

$$W_t^{0*} = G(\mathbf{X}_t, \hat{\beta}_{0n}, \mathbf{W}_t^{L0*}, \hat{\phi}_n) + \varepsilon_t^*, \quad t = p + 1, p + 2, \dots, n,$$

where $\hat{\beta}_{0n} = (\theta_0, \hat{\nu}_n)$ and $\{\varepsilon_t^*\}$ are i.i.d with d.f. F_n , the e.d f. of $\{\hat{\varepsilon}_t\}$.

- (v) Define $C_n(\alpha; \mathbf{Y}_n, \chi_n)$ by

$$P^*(T_n(\mathbf{W}_n^{0*}, \chi_n) \geq C_n(\alpha; \mathbf{Y}_n, \chi_n)) \cong \alpha,$$

where

$$\mathbf{W}_n^{0*} = (W_1^{0*}, W_2^{0*}, \dots, W_n^{0*})^T.$$

Remarks

- (a) There are various ways of obtaining the initial values (when working with lagged variables) $\{W_t^{0*}, t = 1, 2, \dots, p\}$ (in step (iv) above). The simplest and most commonly used method is to set $\{W_t^{0*}, t = 1, 2, \dots, p\}$ equal to the original values $\{Y_t, t = 1, 2, \dots, p\}$, i.e. $W_t^{0*} = Y_t, t = 1, 2, \dots, p$.
- (b) Another method, which is probably more suitable for hypothesis testing as it allows one to mimic H_0 in the bootstrap world more accurately, is to use the so-called “burn-in” period (see, e.g. Davison and Hinkley, 1997). This procedure entails generating $n + k + 1$ bootstrap residuals to obtain $\varepsilon_{-k}^*, \varepsilon_{-k+1}^*, \dots, \varepsilon_0^*, \dots, \varepsilon_n^*$ (the number k is usually chosen to be a fairly

large number). Set

$$\begin{aligned}
 W_{-k}^{0*} &= \varepsilon_{-k}^* \\
 W_{-k+1}^{0*} &= G(\mathbf{X}_t, \hat{\beta}_{0n}, W_{-k}^{0*}, \hat{\phi}_n) + \varepsilon_{-k+1}^* \\
 &\vdots \\
 W_0^{0*} &= G(\mathbf{X}_t, \hat{\beta}_{0n}, W_{-1}^{0*}, \hat{\phi}_n) + \varepsilon_0^* \\
 &\vdots \\
 W_n^{0*} &= G(\mathbf{X}_t, \hat{\beta}_{0n}, W_{n-1}^{0*}, \hat{\phi}_n) + \varepsilon_n^*.
 \end{aligned}$$

In the end, the terms $W_{-k}^{0*}, \dots, W_0^{0*}$ are discarded.

- (c) In (11) we obtained the residuals under the assumption of homoscedastic errors, i.e. $E(\varepsilon_t^2) = \sigma^2$ for all t . If, however, we work with models that are heteroscedastic, i.e. $E(\varepsilon_t^2) = \sigma_t^2$, $t = 1, 2, \dots, n$, bootstrap samples must be generated in such a way that the relationship between the variance of each error term and the corresponding regressors is retained. One possible way to do this is to use the wild bootstrap proposed by Wu (1986), Liu (1988) and further developed by Mammen (1993).

The wild bootstrap is implemented by first obtaining the residuals in (11), without centring them. The bootstrap residuals $\{\varepsilon_t^*, t = 1, 2, \dots, n\}$ are then obtained from a distribution with probability mass in two points. That is

$$\varepsilon_t^* = \begin{cases} a, & \text{with probability } p \\ b, & \text{with probability } 1 - p, \end{cases}$$

where a, b and p are chosen such that $E_*(\varepsilon_t^*) = 0$, $E_*(\varepsilon_t^*)^2 = \hat{\varepsilon}_t^2$ and $E_*(\varepsilon_t^*)^3 = \hat{\varepsilon}_t^3$. One possible choice for a, b and p suggested by Mammen (1993) is $a = \hat{\varepsilon}_t(1 - \sqrt{5})/2$, $b = \hat{\varepsilon}_t(1 + \sqrt{5})/2$ and $p = (\sqrt{5} + 1)/(2\sqrt{5})$. Another possible choice for the two-point distribution is the Rademacher distribution. For this distribution $a = \hat{\varepsilon}_t$, $b = -\hat{\varepsilon}_t$ and $p = \frac{1}{2}$. The reason why it is not necessary to centre the residuals $\{\hat{\varepsilon}_t, t = 1, 2, \dots, n\}$ is because the two-point distribution already has mean 0 (in the bootstrap world). After obtaining the bootstrap residuals $\{\varepsilon_t^*, t = 1, 2, \dots, n\}$ using the wild bootstrap, one can proceed with the residual-based bootstrap algorithm as previously described.

- (d) Another way to deal with heteroscedasticity, which was originally proposed by Freedman (1981), is called *bootstrapping pairs* or the *pairs bootstrap*. Unfortunately, in a hypothesis testing setup, this method of

dealing with heteroscedasticity will not work, as it does not allow one to mimic H_0 in the bootstrap world.

Next, we shall consider an example, namely the location-scale model, where residuals are resampled in order to perform the bootstrap hypothesis test.

Example 2 Location-scale model

Let $\mathbf{X}_n = (X_1, X_2, \dots, X_n)$ denote a random sample from an unknown univariate d.f. F with mean μ and variance σ^2 , $0 < \sigma^2 < \infty$.

Consider the model

$$X_t = \mu + \sigma \varepsilon_t, \quad t = 1, 2, \dots, n,$$

where $\{\varepsilon_t\}$ are i.i.d. random errors with zero mean and variance 1.

Suppose we wish to test the following hypothesis by making use of the bootstrap:

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_A : \sigma^2 > \sigma_0^2.$$

Applying the residual-based bootstrap algorithm, we can obtain the bootstrap critical value $C_{n,P}^R(\alpha; \mathbf{X}_n)$ as follows:

(i) Fit the model and obtain the centred residuals:

$$\hat{\varepsilon}_t = \frac{X_t - \bar{X}_n}{S_n(\mathbf{X}_n)}. \quad (12)$$

(ii) Under H_0 , X_t can be represented as

$$W_t^0 = \mu + \sigma_0 \varepsilon_t^0.$$

(iii) Under H_0^* , W_t^0 can be represented as

$$W_t^{0*} = \bar{X}_n + \sigma_0 \varepsilon_t^*,$$

where $\{\varepsilon_t^*\}$ are i.i.d. bootstrap residuals drawn from the e.d.f. of $\{\hat{\varepsilon}_t\}$ (given in (12)).

(iv) Define $C_{n,P}^R(\alpha; \mathbf{X}_n)$ by

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{W}_n^{0*}) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{W}_n^{0*}) - S_n^4(\mathbf{W}_n^{0*})}} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha. \quad (13)$$

It is easy to see that (13) reduces to

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - S_n^2(\mathbf{X}_n))}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} \geq C_{n,P}^R(\alpha; \mathbf{X}_n) \right) \cong \alpha,$$

which is indeed identical to the expression in (2). If, however, we obtain our residuals in (12) by imposing the null hypothesis, i.e.

$$\hat{\varepsilon}_t = \frac{X_t - \bar{X}_n}{\sigma_0},$$

it is a simple matter to verify that (13) becomes

$$P^* \left(\frac{\sqrt{n}(S_n^2(\mathbf{X}_n^*) - \sigma_0^2)}{\sqrt{\hat{\mu}_4(\mathbf{X}_n^*) - S_n^4(\mathbf{X}_n^*)}} \geq C_{n,P}^W(\alpha; \mathbf{X}_n) \right) \cong \alpha, \quad (14)$$

which is identical to the expression given in (3). It was shown in Theorem 1 that using the critical value defined in (14) leads to a bootstrap test that performs very poorly. In the literature there are a number of authors who choose to resample residuals derived from a restricted model, i.e. where the null hypothesis is imposed when calculating the residuals (see, e.g., Nankervis and Savin, 1996; Park, 2003; Davidson and MacKinnon, 2007). However, many authors, including Van Giersbergen and Kiviet (2002) and Martin (2007), have advocated that resampling must be done from unrestricted residuals (i.e. the null hypothesis must not be imposed when calculating the residuals).

5. Monte Carlo study

In this section we present the results of a Monte Carlo study where we compare the estimated sizes of (R,N-P), (R,P), (W,N-P), (W,P), (T,N-P) and (T,P) for sample sizes $n = 30, 50, 100$ and 200 , respectively. As before, R and W will refer to the wrong and right critical values, respectively, whereas P and N-P refer to the pivotal and non-pivotal cases, respectively. T will refer to the test where the bootstrap critical value is obtained by using the tilted bootstrap. A nominal significance level of $\alpha = 0.05$ was used. Bootstrap critical values

were based on $B = 1000$ independent replications. All calculations were done using double precision arithmetic in FORTRAN and routines from the IMSL library.

Data were generated from three different distributions:

- Normal distribution with mean 0 and variance 1;
- Double exponential distribution with mean 0 and variance 2;
- Exponential distribution with mean 1 and variance 1.

Size estimates were calculated as the proportion of 4 000 Monte Carlo samples that resulted in rejection of H_0 using the bootstrap critical values $\hat{C}_{n,N-P}^R(\alpha; \mathbf{X}_n)$, $\hat{C}_{n,P}^R(\alpha; \mathbf{X}_n)$, $\hat{C}_{n,N-P}^W(\alpha; \mathbf{X}_n)$, $\hat{C}_{n,P}^W(\alpha; \mathbf{X}_n)$, $\hat{C}_{n,N-P}^T(\alpha; \mathbf{X}_n)$ and $\hat{C}_{n,P}^T(\alpha; \mathbf{X}_n)$, respectively. The standard errors of the estimated sizes are less than or equal to $\sqrt{0.25/4\,000} = 0.0079$.

Table 1. Estimated size of bootstrap hypothesis tests for the variance

Distribution	n	(R, N-P)	(R, P)	(W, N-P)	(W,P)	(T, N-P)	(T, P)
Normal	30	0.089	0.046	0.000	0.000	0.060	0.054
	50	0.078	0.047	0.000	0.000	0.055	0.053
	100	0.069	0.051	0.000	0.000	0.054	0.053
	200	0.058	0.050	0.000	0.000	0.051	0.052
Double exponential	30	0.092	0.031	0.000	0.000	0.063	0.041
	50	0.075	0.036	0.000	0.000	0.059	0.044
	100	0.061	0.041	0.000	0.000	0.057	0.045
	200	0.055	0.045	0.000	0.000	0.055	0.046
Exponential	30	0.099	0.020	0.000	0.000	0.072	0.060
	50	0.082	0.029	0.000	0.000	0.071	0.058
	100	0.066	0.037	0.000	0.000	0.065	0.055
	200	0.054	0.046	0.000	0.000	0.062	0.052

From Table 1 we can conclude the following:

- (a) The estimated sizes of (W,P) and (W,N-P) are 0 even for large sample sizes. This agrees with the result derived in Theorem 1. The poor performance of these tests again confirms the importance of applying the bootstrap correctly in hypothesis testing, as highlighted by Hall and

Wilson (1991) in their first guideline.

- (b) In terms of estimated sizes, it is clear that (R,P) outperforms (R,N-P) and that (T,P) also outperform (T,N-P). This agrees with the second guideline given by Hall and Wilson (1991).
- (c) The estimated sizes of (T,P) are in most of the cases, except for the normal distribution, closer to the nominal significance level than those of (R,P), especially for smaller sample sizes.
- (d) Schenker (1985) discussed the problem of constructing bootstrap confidence intervals based on $T_{n,N-P}(\mathbf{X}_n)$ for the variance of a population and showed that the coverage errors of these intervals were unsatisfactorily large. On the other hand, the performance of the (R,P) and (T,P) tests generally yielded satisfactory probability of type I errors.

6. Concluding remarks

It is clear from the discussions in this paper, that it is preferable to make use of a pivotal test statistic and that one must resample from “data” that mimic H_0 . These findings are in agreement with the two guidelines proposed by Hall and Wilson (1991). Although the tilted bootstrap produced better results for the example we considered in this paper, it has some drawbacks, which makes it less attractive for the practical user: it is computationally more intensive to execute than the other methods and a solution for the Lagrange multiplier λ cannot always be obtained, especially in small samples. For more complex testing scenarios the transformation method is also much simpler to implement.

Finally, we strongly advise that bootstrap residuals must be sampled (with replacement) from unrestricted residuals in order to apply the bootstrap correctly to hypothesis testing. The importance of this essential guideline is highlighted in the example discussed above and from the results of the Monte Carlo study.

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